

Chapter 8: Path Integral Control

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Abstract. *This section extends the KL control framework to the class of continuous-time, continuous-state stochastic optimal control (SOC) problems. As in the discrete-time setting, this class of SOC problems admits Monte Carlo-based solution algorithms, commonly known as path integral control (PIC). An alternative derivation of the path integral control framework using the Feynman-Kac formula is also presented.*

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8.1 Deriving PIC using the variational formula

We now formulate continuous-time, continuous-state KL control problems. We assume that the state space $\mathcal{X}$ is equal to $\mathbb{R}^n$, although generalizations to state spaces $\mathcal{X} \subset \mathbb{R}^n$ are possible. All random variables and processes are defined on the underlying probability space $(\Omega, \mathcal{F}, P)$.

8.1.1 State Space Model

Let $V_t \in \mathbb{R}^n$ and $W_t \in \mathbb{R}^m$ be mutually independent Brownian motions under P , and suppose $\mathcal{F}_t \subseteq \mathcal{F}$ is a non-decreasing family of σ -algebras such that both V_t and W_t are martingales with respect to $\mathcal{F}_t$. Assume that the state process $X_t \in \mathbb{R}^n$ is an n -dimensional Ito process:

$$dX_t = f_t(X_t) dt + g_t(X_t) dU_t + h_t(X_t) dV_t, \quad (8.1)$$

where $U_t \in \mathbb{R}^m$ is an m -dimensional control input specified by

$$dU_t = \mu_t dt + \nu_t dW_t. \quad (8.2)$$

The coefficients μ_t and ν_t in (8.2) define the control policy to be designed. Substituting (8.2) into (8.1), we obtain the closed-loop system:

$$dX_t = (f_t + g_t \mu_t) dt + g_t \nu_t dW_t + h_t dV_t. \quad (8.3)$$

For (8.3) to define a well-posed Ito process, we require that $f_t + g_t k_t$ satisfy conditions (i)–(iii) in Section 4.3 (page 57), and that $g_t \nu_t$ and h_t satisfy conditions (i)’–(iii)’ (page 57).

The presence of the $\nu_t dW_t$ term in (8.2) allows the policy to be randomized. Indeed, optimal policies for continuous-time KL control problems are generally randomized, as in the discrete-time KL control setting (Chapter 7). This contrasts with the standard SOC frameworks discussed in Chapter 2 (discrete time) and Chapter 5 (continuous time), where optimal policies can be found within the class of deterministic policies (recall Theorem 2.2 and Theorem 5.1).

We now introduce a key assumption needed to formulate and solve the KL control problem in the sequel. This assumption generalizes the “deterministic transition” condition from previous chapters.

Assumption 8.1 (*Deterministic Transition*) We assume $h_t(X_t) = 0$ for all $t \in [0, \infty)$ in (8.1).

Assumption 8.1 states that the state equation (8.1) does not include the process noise term $h_t dV_t$. At first glance, this may appear overly restrictive, since one would generally wish to handle systems with unpredictable disturbances. However, this restriction can be mitigated by “reinterpreting” the closed-loop system (8.3). Even if $h_t = 0$ in (8.3), the system still contains the stochastic term $g_t \nu_t dW_t$, originating from the randomized control policy (8.2). If we *design* the KL control problem so that $g_t \ell_t dW_t$ coincides with the true process noise, the policy-synthesis framework can be reinterpreted as synthesizing a deterministic control law k_t in the presence of disturbance $g_t \ell_t dW_t$.

The path integral control framework developed by Kappen [1] exploits this interpretation (albeit implicitly) to apply algorithms for KL control (namely Monte-Carlo-based synthesis) to solve stochastic optimal control problems that are seemingly different from the problem formulated above. We will examine this mechanism in detail in the sequel.

8.1.2 Cost functional

Now we introduce the cost functional to be minimized. We will focus on the fixed end-time case, but generalizations to variable end-time cases are possible. Recall that in previous chapters the KL control problems were presented in both stage-wise and trajectory-level forms.¹ Since the notion of “stages” does not naturally arise in continuous time, we begin by deriving a trajectory-level representation.

State-Dependent Cost

Let $C_t(x) : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}$. We assume C_t is twice continuously differentiable on $[0, \infty) \times \mathbb{R}^n$ so that $C_t(X_t)$ is a well-defined Ito process. We also introduce a terminal cost $C_T^{\text{exit}} : \mathbb{R}^n \rightarrow \mathbb{R}$, assumed to be Borel measurable. The state-dependent trajectory cost is

$$\mathbb{E} C_{[0,T]}(X_{[0,T]}) = \mathbb{E} \left[\int_0^T C_t(X_t) dt + C_T^{\text{exit}}(X_T) \right]. \quad (8.4)$$

¹Discrete-time, discrete-state KL control problems had the stage-wise form (7.9) and the trajectory-level form (7.13). Discrete-time, continuous-state KL control problems had stage-wise form (7.46) and trajectory-level form (7.49).

In what follows, we will keep using the notation for the path integral:

$$C_{[0,T]}(X_{[0,T]}) := \int_0^T C_t(X_t) dt + C_T^{\text{exit}}(X_T). \quad (8.5)$$

KL-Divergence Cost

Define the nominal control policy by setting $\mu_t = 0$ and $\nu_t = 1$ in (8.2); that is, the nominal input is simply $dU_t = dW_t$. This choice is made for convenience – other nominal policies are possible. We call this nominal policy R . Specifically, under the nominal control policy R , we call the closed-loop dynamics “uncontrolled” or “passive”:

$$R : \quad \begin{cases} dX_t = f_t(X_t) dt + g_t(X_t) dU_t, & X_0 = x_0 \\ dU_t = dW_t, & U_0 = 0. \end{cases} \quad (8.6)$$

Under a general policy “ Q ,” the closed-loop dynamics are

$$Q : \quad \begin{cases} dX_t = f_t(X_t) dt + g_t(X_t) dU_t, & X_0 = x_0 \\ dU_t = \mu_t dt + dW_t. & U_0 = 0. \end{cases} \quad (8.7)$$

Let $(x_{[0,T]}, u_{[0,T]})$ be an observed sample path. The likelihood ratio (Radon-Nikodym derivative) $\frac{dQ}{dR}(x_{[0,T]}, u_{[0,T]})$ of this sample path can be evaluated using the Girsanov theorem (Theorem 4.4).

To apply the Girsanov theorem, one should understand the relationship between Q and R in terms of the change of measures. Specifically, Q is the probability measure in which W_t is the standard Brownian motion, whereas R is the probability measure in which $dU_t = \mu_t dt + dW_t$ is the standard Brownian motion. Then, by Theorem 4.4,

$$\frac{dQ}{dR}(x_{[0,T]}, u_{[0,T]}) = M_T^{-1} = \exp\left(\int_0^T \mu_t^\top dw_t + \frac{1}{2} \int_0^T \|\mu_t\|^2 dt\right). \quad (8.8)$$

This equation can be interpreted as saying that observing a sample path $(x_{[0,T]}, u_{[0,T]})$ under Q is M_T^{-1} times more likely than observing the same path under R .

As before, the KL divergence between Q and R is defined by

$$D(Q\|R) = \int_{\mathcal{X}_{[0,T]} \times \mathcal{U}_{[0,T]}} \left(\log \frac{dQ}{dR}(x_{[0,T]}, u_{[0,T]}) \right) dQ(x_{[0,T]}, u_{[0,T]}). \quad (8.9)$$

Using (8.8), and noticing that $\int_0^T \mu_t^\top dW_t$ is a martingale under Q , the KL divergence can be explicitly calculated as

$$\begin{aligned} D(Q\|R) &= \mathbb{E}^Q \log M_T^{-1} \\ &= \mathbb{E}^Q \left[\int_0^T \mu_t^\top dW_t + \frac{1}{2} \int_0^T \|\mu_t\|^2 dt \right] \\ &= \frac{1}{2} \int_0^T \mathbb{E}^Q \|\mu_t\|^2 dt. \end{aligned} \quad (8.10)$$

Remark 8.1 In (8.7) we already assumed that $\nu_t = 1$ was selected as the diffusion coefficient of the randomized policy. This choice can be made without loss of generality, since if $\nu_t \neq 1$, the Girsanov transformation between Q and R discussed below becomes singular and the KL divergence becomes infinite.

8.1.3 The KL Control Problem

For $\alpha > 0$, the continuous-time, continuous-state KL control problem (with fixed end time) is formulated as

$$\min_Q \left\{ \mathbb{E}^Q \left[\int_0^T C_t(X_t) dt + C_T^{\text{exit}}(X_T) \right] + \alpha D(Q\|R) \right\}. \quad (8.11)$$

This minimization is over the space of measures Q over the path space $\mathcal{X}_{[0,T]} \times \mathcal{U}_{[0,T]}$ such that $Q \sim R$. For a fixed measure R , every $\mathcal{F}_t^U$ -measurable function μ_t defines a $Q \sim R$ via

$$D(Q\|R) = \frac{1}{2} \int_0^T \mathbb{E}^Q \|\mu_t\|^2 dt. \quad (8.12)$$

Conversely, by Theorem 4.5, for any $Q_{X,U}$ such that $Q_{X,U} \sim R_{X,U}$, there exists an $\mathcal{F}_t$ -measurable function μ_t such that (8.12) holds. Therefore, the KL control problem (8.11) can be written equivalently as a minimization problem with respect to an $\mathcal{F}_t$ -measurable function μ_t :

$$\min_{\mu_{[0,T]}} \mathbb{E}^Q \left[\int_0^T C_t(X_t) dt + C_T^{\text{exit}}(X_T) \right] + \alpha D(Q\|R). \quad (8.13)$$

Using (8.12), the KL control problem (8.13) can also be written as

$$\min_{\mu_{[0,T]}} \mathbb{E}^Q \left[\int_0^T \left\{ C_t(X_t) + \frac{\alpha}{2} \|\mu_t\|^2 \right\} dt + C_T^{\text{exit}}(X_T) \right] \quad (8.14)$$

subject to the stochastic differential equation

$$dX_t = (f_t + g_t \mu_t) dt + g_t dW_t. \quad (8.15)$$

This is a continuous-time stochastic optimal control problem with control-affine dynamics (8.15) and a quadratic-in-control cost (8.14). Such problems were studied in Chapter 5, where the corresponding stochastic HJB equation was derived. This observation implies that a certain class of stochastic optimal control problems can be equivalently expressed as KL control problems, even though such an equivalence may not be initially obvious.

In Section 8.1.6, we will identify precisely when a continuous-time SOC problem can be rewritten as a KL control problem. Recognizing this equivalence is valuable in practice: the KL control framework admits Monte Carlo-based solution algorithms that are generally unavailable for arbitrary SOC problems.

8.1.4 Value Function and Its Free-Energy Representation

Define the value function $V : [0, T] \times \mathcal{X} \rightarrow \mathbb{R}$ by

$$V_t(x) \triangleq \inf_Q \left\{ \mathbb{E}^Q \left[\int_t^T C_s(X_s) ds + C_T^{\text{exit}}(X_T) \right] + \alpha D(Q\|R) \right\}. \quad (8.16)$$

Here, Q and R are understood to mean probability measures over the path space $\mathcal{X}_{[t,T]} \times \mathcal{U}_{[t,T]}$. Specifically,

$$Q : \begin{cases} dX_s = f_s(X_s) ds + g_s(X_s) dU_s, & s \in [t, T], X_t = x \\ dU_s = \mu_s ds + dW_s, & s \in [t, T], U_t = 0. \end{cases} \quad (8.17)$$

and

$$R : \begin{cases} dX_s = f_s(X_s) ds + g_s(X_s) dU_s, & s \in [t, T], X_t = x \\ dU_s = dW_s, & s \in [t, T], U_t = 0. \end{cases} \quad (8.18)$$

Equation (8.16) can be written more explicitly as

$$V_t(x) = \inf_Q \int_{\mathcal{X}_{[t,T]} \times \mathcal{U}_{[t,T]}} \left\{ C_{[t,T]}(x_{[t,T]}) + \alpha \log \frac{dQ}{dR}(x_{[t,T]}, u_{[t,T]}) \right\} dQ(x_{[t,T]}, u_{[t,T]}) \quad (8.19)$$

By the variational formula, this minimization problem can be solved explicitly as

$$V_t(x) = -\alpha \log \left\{ \int_{\mathcal{X}_{[t,T]} \times \mathcal{U}_{[t,T]}} \exp \left(-\frac{1}{\alpha} C_{[t,T]}(x_{[t,T]}) \right) dR(x_{[t,T]}, u_{[t,T]}) \right\}$$

The right-hand side is the free energy representation of the value function and can be written more compactly as

$$-\alpha \log \mathbb{E}^R \exp \left(-\frac{1}{\alpha} C_{[t,T]}(X_{[t,T]}) \right). \quad (8.20)$$

This result indicates that the value function can be computed by evaluating the expected value of the exponentiated path cost with respect to the probability measure R .

8.1.5 Solution via Monte-Carlo Simulation

The variational formula (Proposition 6.1) also shows that there exists a probability measure Q^* that achieves the infimum in (8.19). Let B be a Borel set in the space of continuous functions $\mathcal{X}_{[t,T]} \times \mathcal{U}_{[t,T]}$ starting from $(x_t, u_t) = (x, 0)$. The optimal Q^* is characterized as

$$Q^*(B) = \frac{1}{Z} \int_B \exp \left(-\frac{1}{\alpha} C_{[t,T]}(x_{[t,T]}) \right) dR(x_{[t,T]}, du_{[t,T]}) \quad (8.21)$$

where Z is a normalizing constant. Using the indicator function $\mathbb{1}_B$, (8.21) can also be written as

$$Q^*(B) = \frac{\mathbb{E}^R [\mathbb{1}_B(X_{[t,T]}, U_{[t,T]}) \exp (-\frac{1}{\alpha} C_{[t,T]}(X_{[t,T]}))]}{\mathbb{E}^R [\exp (-\frac{1}{\alpha} C_{[t,T]}(X_{[t,T]}))]}. \quad (8.22)$$

The result (8.21) and (8.22) specify the probability distribution of the control input under the optimal policy Q^* . For example, the probability that the control input $U_{t+\Delta t}$ at some future time $t + \Delta t$ with $\Delta t > 0$ takes a value in a set $\mathcal{U}_s \subseteq \mathcal{U}$ is given by (8.22) with B chosen as $B = \{\omega \in \Omega : U_{t+\Delta t}(\omega) \in \mathcal{U}_s\}$.

We now consider how a control input U_t can be computed according to the optimal policy Q^* in real-time. Since a truly continuous-time implementation of Q^* is not possible on digital computer devices, in what follows, we present a time-discretized implementation of Q^* . In the presented implementation, the control input is updated only at discrete time instances $\tau = t, t + \Delta t, \dots, T - \Delta t$, and is kept constant over the interval $(\tau, \tau + \Delta t]$. At every time step τ , a batch of Monte-Carlo simulations is performed to compute the value of control input u_τ which is held constant during the subsequent time interval $(\tau, \tau + \Delta t]$.

Sampling by the inverse transform method

We first observe that sample trajectories of the future control input $U_s, s \geq t$ under the optimal policy Q^* can be drawn by a Monte-Carlo algorithm similar to Algorithm 14 in Section 7.2.4 (page 121). Specifically, let $\{x_{[t,T]}(i), u_{[t,T]}(i)\}_{i=1}^N$ be sample trajectories generated independently under the reference measure R with the current state x_t as the initial condition. For each i , let

$$r(i) = \exp\left(-\frac{1}{\alpha} C_{[t,T]}(x_{[t,T]}(i))\right) \quad (8.23)$$

be the reward of the sample path i . Set $r = \sum_{i=1}^N r(i)$. Define a monotonically non-decreasing function $F : [0, N] \rightarrow [0, r]$ by

$$F(x) = \sum_{i=1}^{\lfloor x \rfloor} r(i)$$

and its inverse $F^{-1} : [0, r] \rightarrow \{1, 2, \dots, N\}$. To obtain a control input $u_{[t,T]}$ to implement, simply select the i -th realization from the ensemble $\{u_{[t,T]}(i)\}_{i=1}^N$, where $i = F^{-1}(d)$ and d is a random variable sampled from $d \sim \text{unif}[0, r]$. The correctness of this algorithm follows from Claim 6.1 in Section 6.4

Once a sample trajectory of $u_{[t,T]}$ is obtained, one can adopt it as control input over the time interval $(t, t + \Delta t]$. However, since $u_{[t,T]}$ becomes obsolete by the time $t + \Delta t$, one needs to reconstruct a new control input trajectory $u_{[t+\Delta t, T]}$ by regenerating a new trajectory ensemble $\{x_{[t+\Delta t, T]}(i), u_{[t+\Delta t, T]}(i)\}_{i=1}^N$ starting from the current state $x_{t+\Delta t}$ as the initial condition. One can then implement the control input for the time interval $[t + \Delta t, t + 2\Delta t)$ and so on. Although such a periodic implementation is suboptimal (the performance degrades if Δt is too large), we expect a near-optimal performance if Δt is sufficiently small.

In this scheme, notice that the i -th sample trajectory is selected with probability $r(i)/r$. Thus, the expected value of the control input $U_{t+\Delta t}$ is computed as

$$\mathbb{E}U_{t+\Delta t} = \sum_{i=1}^N \frac{r(i)}{r} u_{t+\Delta t}(i). \quad (8.24)$$

Discrete-time implementation

The method presented above is not yet fully practical since Monte-Carlo simulations of (8.6) under the nominal policy R are assumed to be executed in continuous time. The implementation above also requires to apply a continuous time signal $U_{(t, t+\Delta t]}$ as control input. These are inconvenient features as continuous times signals cannot even be stored in digital computers. To design a more practical scheme, it is reasonable to simulate a time-discretized version of (8.6). By the Euler-Maruyama method, a sample path $x_{[t,T]}$ under (8.6) can be obtained by simulating a discrete time process

$$\hat{X}_{\tau+\Delta t} = \hat{X}_\tau + f_\tau(\hat{X}_\tau)\Delta t + g_\tau(\hat{X}_\tau)\Delta W_\tau, \quad \tau = t, t + \Delta t, \dots, T - \Delta t \quad (8.25)$$

with $\hat{X}_t = x_t$ and $\Delta W_\tau \sim \mathcal{N}(0, \Delta t I)$ are i.i.d. random vectors. For simplicity, we pick Δt such that $\mathcal{T} = (0, \Delta t, 2\Delta t, \dots, T - \Delta t, T)$ is an equi-spaced partition and $t \in \mathcal{T}$. Denote by

$\{\hat{x}_{t:T}(i)\}_{i=1}^N$ the ensemble of N (discrete time) sample paths generated by (8.25) and let

$$\hat{r}_t(i) = \exp \left(-\frac{1}{\alpha} \sum_{\tau=t}^{T-\Delta t} C_\tau(\hat{x}_\tau(i)) \Delta t - \frac{1}{\alpha} C_T^{\text{exit}}(\hat{x}_T(i)) \right) \quad (8.26)$$

be the corresponding path reward. As before, we set $\hat{r}_t = \sum_{i=1}^N \hat{r}_t(i)$.

To simplify the control implementation, we consider obtaining a Monte-Carlo estimate $\hat{\mu}_t$ of the optimal μ_t in (8.17) only at discrete time instances $\tau = t, t + \Delta t, \dots, T$, and hold the value constant over the time interval $(\tau, \tau + \Delta t]$. In this case, the integration of (8.17) over the time interval $(\tau, \tau + \Delta t]$ gives us

$$U_{t+\Delta t} = u_t + \hat{\mu}_t \Delta t + \Delta W_t, \quad \Delta W_t \sim \mathcal{N}(0, \Delta t). \quad (8.27)$$

Taking the expectation on both sides, we have

$$\mathbb{E}U_{t+\Delta t} = u_t + \hat{\mu}_t \Delta t. \quad (8.28)$$

To choose $\hat{\mu}_t$ optimally, we apply (8.24) to evaluate the right hand side of (8.28). By the law of large numbers, one can expect

$$\mathbb{E}U_{t+\Delta t} \approx \sum_{i=1}^N \frac{\hat{r}_t(i)}{\hat{r}_t} \hat{U}_{t+\Delta t}(i) \quad (8.29)$$

for an N sufficiently large. Substituting (8.29) into (8.28), $\hat{\mu}_t$ can be estimated as

$$\hat{\mu}_t \approx \frac{1}{\Delta t} \left(\sum_{i=1}^N \frac{\hat{r}_t(i)}{\hat{r}_t} \hat{U}_{t+\Delta t}(i) - u_t \right). \quad (8.30)$$

However, since $\hat{U}_{t+\Delta t}(i) = u_t + \Delta W_t(i)$ by construction (8.25), (8.30) is simplified to

$$\hat{\mu}_t \approx \frac{1}{\Delta t} \left(\sum_{i=1}^N \frac{\hat{r}_t(i)}{\hat{r}_t} (u_t + \Delta W_t(i)) - u_t \right) \quad (8.31a)$$

$$= \sum_{i=1}^N \frac{\hat{r}_t(i)}{\hat{r}_t} \frac{\Delta W_t(i)}{\Delta t}. \quad (8.31b)$$

The quantity (8.31b) is easy to compute and thus is suitable for discrete-time implementations. This leads to a practical implementation of the path integral control algorithm, as summarized in Algorithm 15.

8.1.6 KL control vs. general SOC

In Section 8.1.3, we saw that the SOC problem

$$\min_{\mu_{[0,T]}} \mathbb{E} \left[\int_0^T \left\{ C_t(X_t) + \frac{\alpha}{2} \|\mu_t\|^2 \right\} dt + C_T^{\text{exit}}(X_T) \right] \quad (8.32)$$

Algorithm 15: Path integral control: Monte-Carlo-based policy synthesis for continuous-time KL control (8.11)

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1 Given: Initial state  $x_0$ ;
2 Select a discretization step size  $\Delta t$  and obtain a discrete-time model (8.25);
3 for  $t = 0, 1, \dots, T - \Delta t$  do
4   for  $i = 1, 2, \dots, N$  do
5     Generate a sample path  $\{x_{\tau:T}(i), u_{\tau:T-\Delta t}(i)\}$  starting from  $x_k$  driven by an i.i.d.
       noise process  $\Delta W_\tau \sim \mathcal{N}(0, \Delta t I)$ ,  $\tau = t, t + \Delta t, \dots, T - \Delta t$  using (8.25);
6     For each  $i = 1, 2, \dots, N$ , compute the path reward  $\hat{r}(i)$  by (8.26).
7   Compute  $\hat{r} = \sum_{i=1}^N \hat{r}(i)$ ;
8   Select a control input as  $\mu_t \leftarrow \hat{\mu}_t = \sum_{i=1}^N \frac{\hat{r}_t(i)}{\hat{r}_t} \frac{\Delta W_t(i)}{\Delta t}$ ;
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subject to the state space model

$$dX_t = f_t dt + g_t dU_t \quad (8.33a)$$

$$dU_t = \mu_t dt + dW_t \quad (8.33b)$$

is equivalent to the KL control problem (8.11). An important implication of this observation is that the path integral control algorithm (Algorithm 15) is directly applicable to this class of SOC problems.² This leads us to the following question: Under what conditions can a given SOC problem be reformulated as an equivalent KL control problem?

To answer this, let us revisit the general class of SOC problems considered in Chapter 5. Consider the SOC problem

$$\min_{u_{[0,T]}} \mathbb{E} \left[\int_0^T (C_t(X_t) + \frac{1}{2} u_t^\top R_t(X_t) u_t) dt + \psi(X_T) \right], \quad (8.34)$$

subject to the dynamics

$$dX_t = (f_t(X_t) + g_t(X_t) u_t) dt + \sigma_t(X_t) dW_t, \quad t \in [0, T], \quad (8.35)$$

with $X_0 = x_0 \in \mathbb{R}^n$. Without loss of generality, we assume that $\sigma_t(x)$ has full column rank; that is, there exists a left inverse $\sigma_t(x)^\dagger$ such that $\sigma_t(x)^\dagger \sigma_t(x) = I$. The question posed above can now be answered as follows:

Theorem 8.1 *Suppose there exists a positive constant $\alpha > 0$ such that*

$$\sigma_t(x) \sigma_t(x)^\top = \alpha g_t(x) R_t(x)^{-1} g_t(x)^\top \quad (8.36)$$

holds for all (x, t) . Then the SOC problem (8.34) can be rewritten as (8.32).

Remark 8.2 *Theorem 8.1 shows that an SOC problem of the form (8.34) satisfying the additional condition (8.36) admits an equivalent KL control formulation. Consequently, the path integral control algorithm (Algorithm 15) becomes applicable to this class of SOC problems.*

²By “directly applicable,” we mean that the control input k_t computed by Algorithm 15 becomes optimal in the limit $\Delta t \rightarrow 0$ and $N \rightarrow \infty$.

Proof: In what follows, the parameters $g_t(x) \in \mathbb{R}^{n \times m}$, $\sigma_t(x) \in \mathbb{R}^{n \times m}$, $R_t(x) \in \mathbb{R}^{m \times m}$ are simply denoted as g , σ and R .

Let $\bar{\sigma} \in \mathbb{R}^{n \times (n-m)}$ be a matrix such that $\Sigma = \begin{bmatrix} \sigma & \bar{\sigma} \end{bmatrix} \in \mathbb{R}^{n \times n}$ is non-singular and $\sigma^\top \bar{\sigma} = 0$. Let $\Sigma^{-1} = \begin{bmatrix} \sigma^\dagger \\ \bar{\sigma}^\dagger \end{bmatrix}$ be the inverse of Σ satisfying both

$$\begin{bmatrix} \sigma^\dagger \\ \bar{\sigma}^\dagger \end{bmatrix} \begin{bmatrix} \sigma & \bar{\sigma} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \text{ and } \begin{bmatrix} \sigma & \bar{\sigma} \end{bmatrix} \begin{bmatrix} \sigma^\dagger \\ \bar{\sigma}^\dagger \end{bmatrix} = I. \quad (8.37)$$

Since $\bar{\sigma}^\dagger \sigma = 0$, it follows from (8.36) that $\bar{\sigma}^\dagger g = 0$. Multiplying (8.35) by σ^{-1} from the left, we obtain

$$\begin{bmatrix} \sigma^\dagger \\ \bar{\sigma}^\dagger \end{bmatrix} dX_t = \begin{bmatrix} \sigma^\dagger f \\ \bar{\sigma}^\dagger f \end{bmatrix} dt + \begin{bmatrix} \sigma^\dagger g \\ 0 \end{bmatrix} u_t dt + \begin{bmatrix} I \\ 0 \end{bmatrix} dw_t. \quad (8.38)$$

Setting $k_t := \sigma^\dagger g u_t$, we have

$$\begin{bmatrix} \sigma^\dagger \\ \bar{\sigma}^\dagger \end{bmatrix} dX_t = \begin{bmatrix} \sigma^\dagger \\ \bar{\sigma}^\dagger \end{bmatrix} f dt + \begin{bmatrix} I \\ 0 \end{bmatrix} (k_t dt + dw_t). \quad (8.39)$$

Multiplying (8.39) by $\Sigma = \begin{bmatrix} \sigma & \bar{\sigma} \end{bmatrix}$ from the left, we obtain

$$dX_t = f dt + \sigma dU_t \quad (8.40)$$

where $dU_t = k_t dt + dw_t$. Therefore, the state space model (8.35) can be rewritten in the form (8.33).

Pre- and post-multiplying (8.36) by $\sigma^\dagger$ and $(\sigma^\dagger)^\top$, we obtain

$$I = (\sqrt{\alpha} \sigma^\dagger g R^{-\frac{1}{2}}) (\sqrt{\alpha} \sigma^\dagger g R^{-\frac{1}{2}})^\top. \quad (8.41)$$

This implies that $\sqrt{\alpha} \sigma^\dagger g R^{-\frac{1}{2}}$ is an orthonormal matrix. Hence,

$$I = (\sqrt{\alpha} \sigma^\dagger g R^{-\frac{1}{2}})^\top (\sqrt{\alpha} \sigma^\dagger g R^{-\frac{1}{2}}) \quad (8.42)$$

$$= \alpha R^{-\frac{1}{2}} g^\top (\sigma^\dagger)^\top \sigma^\dagger g R^{-\frac{1}{2}} \quad (8.43)$$

from which we obtain $R = \alpha g^\top (\sigma^\dagger)^\top \sigma^\dagger g$. In particular,

$$u_t^\top R u_t = \alpha u_t^\top g^\top (\sigma^\dagger)^\top \sigma^\dagger g u_t = \alpha \|k_t\|^2. \quad (8.44)$$

Therefore, the cost functions (8.34) and (8.32) are equivalent. $\blacksquare$

Theorem 8.1, combined with the result (8.20) from Section 8.1.4, implies that under the condition (8.36), the value function $V_t(x)$ for the SOC problem (8.34) admits the free-energy representation

$$V_t(x) = -\alpha \log Z_t(x) \quad (8.45a)$$

$$Z_t(x) = \mathbb{E}^Q \left[\exp \left(-\frac{1}{\alpha} C_{[t,T]}(X_{[t,T]}) \right) \middle| X_t = x \right] \quad (8.45b)$$

$$C_{[t,T]}(X_{[t,T]}) = \int_t^T C_s(X_s) ds + \psi(X_T) \quad (8.45c)$$

where X_s , $t \leq s \leq T$ satisfies the “uncontrolled” dynamics

$$dX_t = f_t(X_t) dt + \sigma_t(X_t) dW_t. \quad (8.46)$$

8.2 Deriving PIC using the Feynman-Kac formula

In this section, we present an alternative derivation of the path integral control algorithm via the linearized HJB equation and the Feynman-Kac formula.

Recall that the discrete-time KL control problem in Chapter 7 was also called the linearly solvable MDPs because, under the assumption that the state transition is deterministic, the associated Bellman equation can be linearized via the change of variables:

$$V_t(x) = -\alpha \log Z_t(x). \quad (8.47)$$

It is natural to speculate that the “linearly solvable” nature generalizes to the continuous-time formulations.

To see that this is indeed the case, let us now study the property of the stochastic HJB equation associated with the KL control problem (8.13). Since the KL control problem (8.13) is equivalent to the SOC problem (8.14) of the standard form, it is reasonable to revisit the standard framework of continuous-time SOC considered in Chapter 5 and study the conditions under which the associated HJB equation can be linearized.

Consider once again the SOC problem (8.34). Theorem 5.1 (verification theorem) states that if there exists a solution $V(x, t)$ that is continuously differentiable in t and twice continuously differentiable in x satisfying the nonlinear partial differential equation (HJB PDE):

$$-\partial_t V = -\frac{1}{2}(\partial_x V)^\top g_t R_t^{-1} g_t^\top (\partial_x V) + C_t + f_t^\top \partial_x V + \frac{1}{2} \text{Tr}(\sigma_t \sigma_t^\top \partial_x^2 V) \quad (8.48a)$$

$$V(x, T) = \psi(x), \quad x \in \mathbb{R}^n \quad (8.48b)$$

over the domain $(x, t) \in \mathbb{R}^n \times [0, T)$, then $V(x, t)$ serves as the value function for (8.34). Assuming such a solution exists, let us apply the change of variable (8.47). We obtain a PDE in terms of $Z(x, t)$:

$$-\partial_t Z = f_t^\top \partial_x Z - \frac{C_t}{\alpha} Z + \frac{1}{2} \text{Tr}(\sigma_t \sigma_t^\top \partial_x^2 Z) + \frac{1}{2Z} (\partial_x Z)^\top (\alpha g_t R_t^{-1} g_t^\top - \sigma_t \sigma_t^\top) (\partial_x Z) \quad (8.49a)$$

$$Z(x, T) = \exp\left(-\frac{\psi(x)}{\alpha}\right), \quad x \in \mathbb{R}^n \quad (8.49b)$$

over the domain $(x, t) \in \mathbb{R}^n \times [0, T)$.

Exercise 8.1 Derive (8.49). *Hint: (8.47) implies*

$$\partial_t V = -\frac{\alpha}{Z} \partial_t Z, \quad \partial_x V = -\frac{\alpha}{Z} \partial_x Z \quad (8.50a)$$

$$\partial_x^2 V = \frac{\alpha}{Z^2} (\partial_x Z)(\partial_x Z)^\top - \frac{\alpha}{Z} (\partial_x^2 Z). \quad (8.50b)$$

The PDE (8.49a) is still nonlinear because the last term is quadratic in $\partial_x Z$. However, the quadratic term vanishes under the following assumption:

Assumption 8.2 For each $(x, t) \in \mathbb{R}^n \times [0, T)$, the condition

$$\sigma_t(x) \sigma_t(x)^\top = \alpha g_t(x) R_t(x)^{-1} g_t(x)^\top \quad (8.51)$$

holds.

Assumption 8.2 imposes a specific relationship among the coefficient g_t , the diffusion coefficient σ_t , and the control cost function R_t . The condition (8.51) is often interpreted as “in a direction with low noise, control is expensive” [1]. It is also worth noting that (8.51) is equivalent to the premise (8.36) of Theorem 8.1.

Under Assumption 8.2, the PDE (8.49a) simplifies to

$$-\partial_t Z = f_t^\top \partial_x Z - \frac{C_t}{\alpha} Z + \frac{1}{2} \text{Tr}(\sigma_t \sigma_t^\top \partial_x^2 Z), \quad \forall (x, t) \in \mathbb{R}^n \times [0, T] \quad (8.52a)$$

$$Z(x, T) = \exp\left(-\frac{\psi(x)}{\alpha}\right), \quad \forall x \in \mathbb{R}^n \quad (8.52b)$$

The linearized PDE (8.52) can be viewed as a continuous-time analogue of the linearized Bellman equation for discrete-time KL control problems derived in Chapter 7.

When a PDE is solved numerically, linearity offers several computational advantages. The following theorem plays a crucial role in connecting the linear PDE (8.52) to Monte Carlo algorithms:

Theorem 8.2 (*The Feynman-Kac formula*) *Let $f_t(x)$, $\sigma_t(x)$, $h_t(t)$, $\ell_t(x)$, $t \in [0, T)$, and $L_T(x)$ be given functions. Suppose that there exists a continuous function $Z : \mathbb{R}^n \times [0, T] \rightarrow \mathbb{R}$ such that*

(a) *$Z(x, t)$ is continuously differentiable in t and twice continuously differentiable in x on $\mathbb{R}^n \times [0, T]$;*

(b) *$Z(x, t)$ satisfies the PDE:*

$$-\partial_t Z = f_t^\top \partial_x Z - \ell_t Z + \frac{1}{2} \text{Tr}(\sigma_t \sigma_t^\top \partial_x^2 Z) + h_t, \quad \forall (x, t) \in \mathbb{R}^n \times [0, T] \quad (8.53a)$$

$$Z(x, T) = L_T(x), \quad \forall x \in \mathbb{R}^n. \quad (8.53b)$$

Then, for each $(x, t) \in \mathbb{R}^n \times [0, T)$, $Z(x, t)$ can be expressed as

$$Z(x, t) = \mathbb{E}^Q \left[G_T L_T(X_T) + \int_t^T G_\tau h_\tau(X_\tau) d\tau \middle| X_t = x \right] \quad (8.54)$$

where

- $W_s, t \leq s \leq T$ is the multidimensional standard Brownian motion in Q ;
- $X_s, t \leq s \leq T$ is the Ito process defined by

$$dX_s = f_s(X_s)ds + \sigma_s(X_s)dW_s, \quad X_t = x; \quad (8.55)$$

- $G_s, t \leq s \leq T$ is the process defined by

$$G_s = \exp\left(-\int_t^s \ell_\tau(X_\tau) d\tau\right). \quad (8.56)$$

Proof: Define the process $Y_s, t \leq s \leq T$ by

$$Y_s = G_s Z(X_s, s) + \int_t^s G_\tau h_\tau(X_\tau) d\tau. \quad (8.57)$$

At the boundary times $s = t$ and $s = T$, Y_s satisfies

$$Y_t = Z(x, t) \quad (8.58)$$

$$Y_T = G_T L_T(X_T) + \int_s^T G_\tau h_\tau(X_\tau) d\tau. \quad (8.59)$$

Therefore, if Y_s is a martingale, then

$$Z(x, t) = Y_t = \mathbb{E}[Y_T \mid X_t = x], \quad (8.60)$$

and the proof is complete. It thus remains to show that Y_s is a martingale.

From (8.57),

$$dY_s = G_s dZ + Z dG_s + dG_s \cdot dZ + G_s h_s(X_s) ds. \quad (8.61)$$

Using the Ito formula,

$$\begin{aligned} dZ &= \partial_s Z ds + (\partial_x Z)^\top dX_s + \frac{1}{2} \text{Tr}[(\partial_s^2 Z)(dX_s)(dX_s)^\top] \\ &= \partial_s Z ds + (\partial_x Z)^\top (f_s ds + \sigma_s dW_s) + \frac{1}{2} \text{Tr}[\sigma_s \sigma_s^\top (\partial_x^2 Z)]. \end{aligned} \quad (8.62)$$

It can be verified that G_s satisfies

$$dG_s = -\ell_s(X_s) G_s ds, \quad G_t = 1. \quad (8.63)$$

Then, by Ito's rule,

$$dG_s \cdot dZ = 0. \quad (8.64)$$

Substituting into (8.61),

$$dY_s = G_s \left(\partial_s Z + f_s^\top \partial_x Z + \frac{1}{2} \text{Tr}[\sigma_s \sigma_s^\top (\partial_x^2 Z)] - \ell_s Z + h_s \right) ds + G_s (\partial_x Z)^\top \sigma_s dW_s. \quad (8.65)$$

Since Z satisfies (8.53a), we obtain $dY_s = G_s (\partial_x Z)^\top \sigma_s dW_s$, which shows that Y_s is a martingale. ■

The representation (8.54) is called the *Feynman-Kac representation* or the *path integral representation* of the solution Z of the PDE. To obtain the path integral representation of (8.52), we simply identify

$$\ell_s(x) = \frac{C_s(x)}{\alpha}, \quad L_T(x) = \exp\left(-\frac{\psi(x)}{\alpha}\right), \quad h_t(s) = 0. \quad (8.66)$$

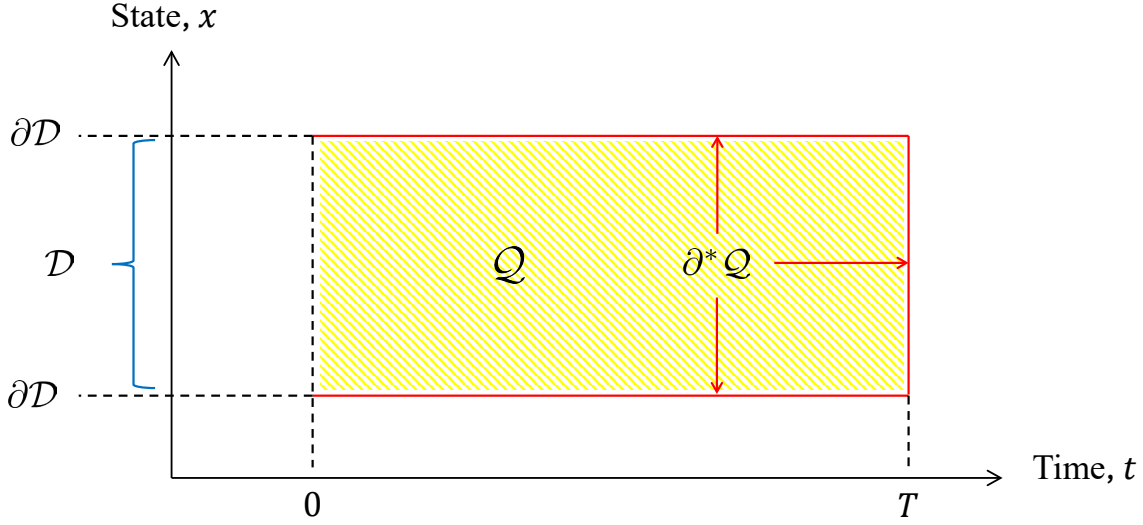
Consequently, we obtain

$$Z(x, t) = \mathbb{E}^Q \left[\exp\left(-\frac{1}{\alpha} C_{[t,T]}(X_{[t,T]})\right) \middle| X_t = x \right] \quad (8.67)$$

where

$$C_{[t,T]}(X_{[t,T]}) = \int_t^T C_s(X_s) ds + \psi(X_T). \quad (8.68)$$

This conclusion is equivalent to our earlier result (8.45).

Figure 8.1: Domain $\mathcal{Q}$ and its parabolic boundary $\partial^* \mathcal{Q}$.

Generalization

It is possible to extend the Feynman-Kac formula to PDEs with more general boundary conditions. Let $\mathcal{D} \subset \mathbb{R}^n$ be an open subset and $\partial\mathcal{D}$ its boundary. Also, let $\mathcal{Q} := [0, T) \times \mathcal{D}$ be the cylindrical region, and define its parabolic boundary as

$$\partial^* \mathcal{Q} = ([0, T) \times \partial\mathcal{D}) \cup (\{T\} \times \mathcal{D}). \quad (8.69)$$

The domain $\mathcal{Q}$ and $\partial^* \mathcal{Q}$ are shown in Figure 8.1.

Let $\psi : \partial^* \mathcal{D} \rightarrow \mathbb{R}$ be a function such that $\psi_t(x)$ is continuous in (x, t) . Suppose that a function $Z : \mathcal{D} \rightarrow \mathbb{R}$ satisfies the following linear PDE with a boundary condition on $\partial^* \mathcal{D}$:

$$-\partial_t Z = f_t^\top \partial_x Z - \ell_t Z + \frac{1}{2} \text{Tr}(\sigma_t \sigma_t^\top \partial_x^2 Z) + h_t, \quad \forall (x, t) \in \mathcal{Q} \quad (8.70a)$$

$$\lim_{(y,s) \rightarrow (x,t)} Z(y, s) = \psi_t(x), \quad \forall (x, t) \in \partial^* \mathcal{Q}. \quad (8.70b)$$

To obtain the Feynman-Kac representation of Z , consider an Ito process

$$dX_s = f_s(X_s)ds + \sigma_s(X_s)dW_s, \quad t \leq s, \quad X_t = x \quad (8.71)$$

where W_s is the standard Brownian motion in the probability measure Q . Let $T_f = \min\{T, \inf\{s > t : X_s \notin \mathcal{D}\}\}$. If $T = \infty$, then T_f is the first exit time. Then it can be shown that

$$Z(x, t) = \mathbb{E}^Q \left[G_{T_f} L_{T_f}(X_{T_f}) + \int_t^{T_f} G_\tau h_\tau(X_\tau) d\tau \middle| X_t = x \right] \quad (8.72)$$

where G_s is defined by (8.56).

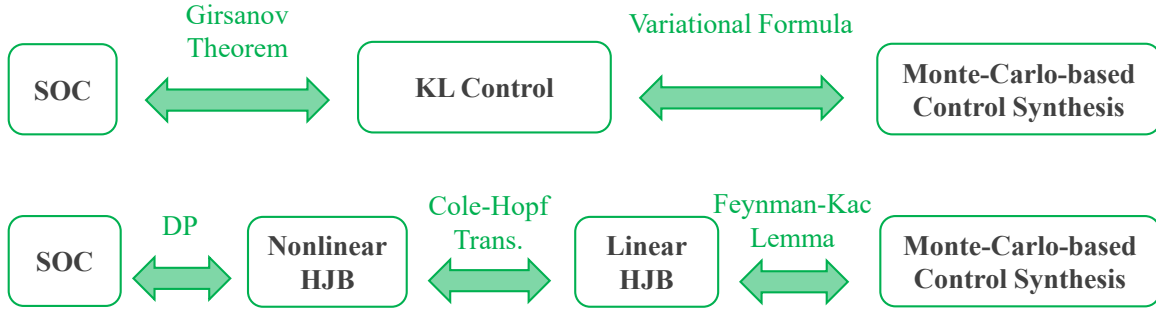


Figure 8.2: Derivation of the path integral control algorithm.

Exercise 8.2 (*Solving PDE by Monte Carlo*) Consider an n -dimensional standard Brownian motion X_t starting from $X_0 = x \in \mathcal{D}$ where $\mathcal{D} = \{x \in \mathbb{R}^n : |x| < R\}$. Let $T_f = \inf\{t > 0 : X_t \notin \mathcal{D}\}$ be the first exit time. Then it can be shown that $Z(x) := \mathbb{E}[T_f | X_0 = x]$ solves the PDE:

$$\frac{1}{2} \text{Tr}(\partial_x^2 Z) + 1 = 0 \quad \forall x \in \mathcal{D} \quad (8.73)$$

$$Z(x) = 0 \quad \forall x \in \partial\mathcal{D}. \quad (8.74)$$

Obtain the Feynman-Kac representation of $Z(x)$. Suggest a Monte Carlo algorithm to compute $Z(x)$. Hint: The result should be very intuitive.

8.3 Discussion

In this chapter, we have developed two different bridges connecting a certain class of continuous-time SOC problems and Monte Carlo-based solution algorithms. Our journey from Chapter 6 to Section 8.1.6 was dedicated to establishing the first bridge (Figure 8.2, top). The approach introduced in Section 8.2 (Feynman-Kac formula) establishes the second bridge (Figure 8.2, bottom). These two approaches are intimately related – for example, both require the same condition ((8.36) and (8.51)) for the SOC problems to be converted into Monte Carlo sampling problems.³

Historically, these two approaches were discovered independently. The foundation of the first approach was Todorov’s linearly solvable MDPs [6, 7]. Kappen [1] independently established the approach via the Feynman-Kac formula. Theodorou and Todorov drew the connection between these two approaches [8]. Yet another derivation of the path integral method is possible by formulating an SOC as a statistical inference problem and then applying standard graphical model inference algorithms [5, 4].

The origin of the path integral control algorithm can be traced back to the stochastic variational treatment of quantum mechanics [9, 10, 11, 12, 13]. Using Nelson’s diffusion model [9], [12, 13] derived an SOC problem whose associated HJB equation coincides with the linear Schrödinger equation. This connection between SOC and quantum mechanics was exploited by Itami [14, 15],

³The relaxation of this assumption has been investigated using KL divergence minimization [2], iterative path integral control [3], and graphical models [4, 5].

who proposed a Monte Carlo (Metropolis) algorithm to evaluate the path integral representation of the wave function. Noticing that the semiclassical limit of the Schrödinger equation approximates the HJB equation, [15] proposed a Monte Carlo-based solution approach to deterministic nonlinear optimal control problems. To the best of our knowledge, this constitutes the earliest appearance of what is now known as path integral control.

Bibliography

- [1] H. J. Kappen, “Path integrals and symmetry breaking for optimal control theory,” *Journal of statistical mechanics: theory and experiment*, vol. 2005, no. 11, p. P11011, 2005.
- [2] G. Williams, N. Wagener, B. Goldfain, P. Drews, J. M. Reh, B. Boots, and E. A. Theodorou, “Information theoretic MPC for model-based reinforcement learning,” in *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2017, pp. 1714–1721.
- [3] S. Satoh, H. J. Kappen, and M. Saeki, “An iterative method for nonlinear stochastic optimal control based on path integrals,” *IEEE Transactions on Automatic Control*, vol. 62, no. 1, pp. 262–276, 2016.
- [4] S. Levine, “Reinforcement learning and control as probabilistic inference: Tutorial and review,” *arXiv preprint arXiv:1805.00909*, 2018.
- [5] H. J. Kappen, V. Gómez, and M. Opper, “Optimal control as a graphical model inference problem,” *Machine learning*, vol. 87, pp. 159–182, 2012.
- [6] E. Todorov, “Linearly-solvable markov decision problems,” *Advances in neural information processing systems*, pp. 1369–1376, 2007.
- [7] —, “Efficient computation of optimal actions,” *Proceedings of the national academy of sciences*, vol. 106, no. 28, pp. 11 478–11 483, 2009.
- [8] E. A. Theodorou and E. Todorov, “Relative entropy and free energy dualities: Connections to path integral and kl control,” *The 51st IEEE Conference on Decision and Control*, 2012.
- [9] E. Nelson, “Derivation of the Schrödinger equation from newtonian mechanics,” *Physical review*, vol. 150, no. 4, p. 1079, 1966.
- [10] H. Rosenbrock, “A variational principle for quantum mechanics,” *Physics letters A*, vol. 110, no. 7-8, pp. 343–346, 1985.
- [11] H. H. Rosenbrock, “A stochastic variational treatment of quantum mechanics,” *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences*, vol. 450, no. 1939, pp. 417–437, 1995.
- [12] F. Guerra and L. M. Morato, “Quantization of dynamical systems and stochastic control theory,” *Physical review D*, vol. 27, no. 8, p. 1774, 1983.
- [13] K. Yasue, “Stochastic calculus of variations,” *Journal of Functional Analysis*, vol. 41, no. 3, pp. 327–340, 1981.
- [14] T. Itami, “Optimization of nonlinear control systems based on the principle of superposition,” *Transactions of the Society of Instrument and Control Engineers*, vol. 37, no. 3, 2001.
- [15] —, “Nonlinear optimal control via Monte-Carlo evaluation of path integrals,” *Transactions of the Institute of Systems, Control and Information Engineers*, vol. 16, no. 12, 2003.